

Lab02 – Creating a Pod and inspecting it

In this exercise, you will practice the creation of a new Pod in a namespace. Once created, you will inspect it, shell into it and run some operations inside of the container.

1. Create the namespace `ckad-prep`.

```
brahim@Training:~/lab02$ kubectl create namespace ckad-prep
namespace/ckad-prep created
brahim@Training:~/lab02$
brahim@Training:~/lab02$ kubectl get namespace ckad-prep
NAME          STATUS    AGE
ckad-prep     Active   12s
brahim@Training:~/lab02$
brahim@Training:~/lab02$
```

2. In the namespace `ckad-prep`, create a new Pod named `mypod` with the image `nginx:2.3.5`. Expose the port 80.

```
brahim@Training:~/lab02$ kubectl run mypod --image=nginx:2.3.5 --restart=Never --port=80 --namespace=ckad-prep
pod/mypod created
brahim@Training:~/lab02$ kubectl get pods -n ckad-prep
NAME    READY   STATUS              RESTARTS   AGE
mypod   0/1     ImagePullBackOff    0           46s
brahim@Training:~/lab02$
brahim@Training:~/lab02$
```

3. Identify the issue with creating the container. Write down the root cause of issue in a file named `pod-error.txt`.

The list of events of the Pod can give you a deeper insight.

```
brahim@Training:~/lab02$ kubectl events pod mypod -n ckad-prep
LAST SEEN          TYPE      REASON      OBJECT      MESSAGE
4m22s              Normal    Scheduled    Pod/mypod    Successfully assigned ckad-prep/mypod to kube-node2
3m52s (x2 over 4m18s) Normal    BackOff      Pod/mypod    Back-off pulling image "nginx:2.3.5"
3m52s (x2 over 4m18s) Warning   Failed       Pod/mypod    Error: ImagePullBackOff
3m38s (x3 over 4m20s) Normal    Pulling      Pod/mypod    Pulling image "nginx:2.3.5"
3m37s (x3 over 4m18s) Warning   Failed       Pod/mypod    Failed to pull image "nginx:2.3.5": rpc error: code = NotFound desc = failed to pull
and unpack image "docker.io/library/nginx:2.3.5": failed
to resolve reference "docker.io/library/nginx:2.3.5": docker.io/library/nginx:2.3.5:
not found
3m37s (x3 over 4m18s) Warning   Failed       Pod/mypod    Error: ErrImagePull
2m37s              Normal    Scheduled    Pod/mypod    Successfully assigned ckad-prep/mypod to kube-node2
52s (x4 over 2m36s) Normal    Pulling      Pod/mypod    Pulling image "nginx:2.3.5"
50s (x4 over 2m21s) Warning   Failed       Pod/mypod    Failed to pull image "nginx:2.3.5": rpc error: code = NotFound desc = failed to pull
and unpack image "docker.io/library/nginx:2.3.5": failed
to resolve reference "docker.io/library/nginx:2.3.5": docker.io/library/nginx:2.3.5:
not found
50s (x4 over 2m21s) Warning   Failed       Pod/mypod    Error: ErrImagePull
25s (x6 over 2m20s) Warning   Failed       Pod/mypod    Error: ImagePullBackOff
13s (x7 over 2m20s) Normal    BackOff      Pod/mypod    Back-off pulling image "nginx:2.3.5"
brahim@Training:~/lab02$
```

4. Change the image of the Pod to `nginx:1.15.12`, and then list the Pod and ensure that the container is running.

Go ahead and edit the existing Pod. Alternatively, you could also just use the `kubectl set image pod mypod mypod=nginx --namespace=ckad-prep` command.

```
brahim@Training:~/lab02$ kubectl set image pod mypod mypod=nginx:1.15.12 --namespace=ckad-prep
brahim@Training:~/lab02$
brahim@Training:~/lab02$ kubectl get pods -n ckad-prep
NAME      READY   STATUS    RESTARTS   AGE
mypod     1/1     Running   0           112s
brahim@Training:~/lab02$
brahim@Training:~/lab02$
```

5. Log into the container and run the `ls` command. Write down the output. Log out of the container.

```
brahim@Training:~/lab02$ kubectl exec mypod -it --namespace=ckad-prep -- /bin/sh
#
# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
#
# exit
brahim@Training:~/lab02$
brahim@Training:~/lab02$
```

6. Retrieve the IP address of the Pod `mypod`.

```
brahim@Training:~/lab02$ kubectl get pods -o wide -n ckad-prep
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE      NOMINATED NODE   READINESS GATES
mypod     1/1     Running   0           5m47s  10.244.2.36   kube-node2  <none>            <none>
brahim@Training:~/lab02$
brahim@Training:~/lab02$
```

7. Run a temporary Pod in the namespace `ckad-prep` using the image `busybox`, shell into it and run a `wget` command against the `nginx` Pod using port 80.

```
brahim@Training:~/lab02$ kubectl run busybox --image=busybox --rm -it --restart=Never -n ckad-prep -- /bin/sh
If you don't see a command prompt, try pressing enter.
/ #
/ #
/ # wget -O- 10.244.2.36
Connecting to 10.244.2.36 (10.244.2.36:80)
writing to stdout
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
100% |*****| 612 0:00:00 ETA
written to stdout
/ #
```

8. Render the logs of Pod `mypod`.

```
brahim@Training:~/lab02$ kubectl logs mypod -n ckad-prep
10.244.2.37 - - [25/Jul/2023:12:08:23 +0000] "GET / HTTP/1.1" 200 612 "-" "wget" "-"
10.244.2.38 - - [25/Jul/2023:12:09:15 +0000] "GET / HTTP/1.1" 200 612 "-" "wget" "-"
10.244.2.39 - - [25/Jul/2023:12:09:38 +0000] "GET / HTTP/1.1" 200 612 "-" "wget" "-"
brahim@Training:~/lab02$
```

9. Delete the Pod and the namespace.

```
brahim@Training:~/lab02$ kubectl delete pod mypod --namespace=ckad-prep
pod "mypod" deleted
brahim@Training:~/lab02$ kubectl delete namespace ckad-prep
namespace "ckad-prep" deleted

```